

Project 1: Vectors

STAT 133, Spring 2026

Introduction

The purpose of this assignment is to give you the opportunity to work with common operations for R vectors, using R to carry out simple computations, and also writing simple functions. Likewise, this project will allow you to put in practice your publishing skills to your QuartoPub account.

General Instructions

- Use a Quarto Document (.qmd file) for your Project 1.
- Save this file as `project1.qmd`.
- You will have to Render your `qmd` file into an HTML file. In other words, your output format in the YAML header should be `format: html`
- You will have to publish your `qmd` file to your **QuartoPub** account. This was covered in lab-2.
- Assume that your published document will be seen by a potential employer, boss, client/customer, admissions committee, or something like that. In other words, make sure that the content of your project is “understandable”: content is well organized, self descriptive, easy to follow, etc.
- To receive full credit, please submit the following to bCourses (proj1 assignment):
 - `qmd` file
 - (URL) link to your published document

Problems start in the next page

1 Dogs Data

1.1 Image of dogs

Search for an image of dogs and include it (i.e. insert it) in your document (example of image shown below).



1.2 Vectors

Consider the following data about 10 dog breeds

breed	avg_life	energy	trainability	friendly
Beagle	12	high	0.6	TRUE
Bulldog	9	medium	0.6	TRUE
Collie	13	medium	0.8	TRUE
German Shepherd	9	medium	1	FALSE
Golden Retriever	11	very high	1	TRUE
Greyhound	12	high	NA	TRUE
Labrador Retriever	11	very high	1	FALSE
Pomeranian	14	medium	0.6	FALSE
Pug	14	medium	0.6	TRUE
Shiba Inu	15	medium	0.4	FALSE

Create vectors and factors for the columns in the data table displayed above, according to the following data types. Encode missing values as NA. BTW: You **cannot** use any of the explicit coercion functions: e.g. `as.integer()`, `as.logical()`, etc.

- **breed**: character vector
- **avg_life**: integer vector (expectancy life in years)
- **energy**: factor (energy level); *hint*: `factor()` is your friend
- **trainability**: double or real vector (trainability level)
- **friendly**: logical vector (tends to be very friendly)

1.3 Subscripting (aka Indexing, Slicing)

Use **bracket notation** to write R commands—displaying the output—that answer the following questions. Answer each part in its individual code chunk.

- a) What is the breed with the longest average life expectancy diameter? *Hint:* the function `which.max()` is your friend.
- b) How many breeds don't tend to be very friendly? *Hint:* the function `sum()` is your friend.
- c) What is the median trainability value of those breeds that are friendly?.
- d) How many breeds have average life expectancy values smaller than the median average life expectancy?
- e) How many breeds have medium energy level and tend to be very friendly?
- f) What breeds have medium energy level and tend to be very friendly?
- g) What is the minimum trainability value for those breeds with an average lifetime of 13 years or more? *Hint:* Pay attention to NA values.

2 Vectorized Functions

For each function or operator listed below, state whether the function/operator is vectorized and provide a code example to support your claim.

- 1. `length()`
- 2. `*`
- 3. `log()`
- 4. `max()`
- 5. `trunc()`

3 Loan Amortization Payment Formula

Consider the following formula for computing the periodic payment (PMT) of a loan amortization:

$$\text{PMT} = P \times \frac{\frac{r}{k} \left(1 + \frac{r}{k}\right)^{nk}}{\left(1 + \frac{r}{k}\right)^{nk} - 1} \quad (1)$$

where:

- P = principal (lent amount)
- PMT = periodic payment (typically monthly)
- r = interest rate
- n = number of years
- k = number of periodic payments (typically 12: monthly)

What it is: when $k = 12$, this formula gives you the monthly payment of a loan amortization.

How to use it: calculate the fixed periodic—usually monthly—payment of a loan.

Best for: Amortization is used in personal loan, home loan (mortgage), auto loan, student loan repayment schedule preparation.

3.1 Computing Payment

Write R code to answer the following question.

Suppose that one of your friends recently graduated from a College in California with \$45,000 in student loan debt, at an interest rate of 5.25%. What would be the monthly payment if your friend wants to pay-off their student loan in 20 years?

3.2 Computing Years

Write R code to answer the following question.

Your friend has various monthly payments to make such as rent, car loan, cell-phone, internet, streaming services, etc. With so many things to pay for, your friend is thinking of making monthly payments of \$220 to pay for their student loan. How many years would it take your friend to be student-loan debt free if they plan to make monthly payments of \$220? *Hint:* use a trial-and-error process for various values of n to find the answer.

4 Future Value of Ordinary Annuity

Mrs. Savingswood wants to save money for the college education of her 3 year-old daughter. She plans to deposit \$100 at the end of each month, for the next 15 years, into a [mutual fund](#) that gives—on average—a 10% annual rate of return. How much money will Mrs. Savingswood have at the end of 15 years?

We can answer this question with the Future Value of an Ordinary Annuity formula:

$$\text{FV ordinary annuity} \rightarrow \text{FV} = \text{PMT} \times \frac{\left(1 + \frac{r}{k}\right)^{nk} - 1}{r/k} \quad (2)$$

where:

- FV = future value (amount accumulated)
- PMT = periodic contribution made at the end of each period
- r = annual interest rate (i.e. annual rate of return)
- n = number of years
- k = number of compounding periods per year; e.g. yearly ($k = 1$), twice yearly ($k = 2$), quarterly ($k = 4$) and monthly ($k = 12$).

Notice that if $r = 0$ then:

$$\text{FV} = \text{PMT} \times nk$$

4.1 Function `annuity()`

- Based on the formula from equation [2](#), write a function `annuity()`, that allows you to compute the **future value of an ordinary annuity**.
- Make sure your function works with $r = 0$
- Give descriptive names to the arguments of the function.
- Likewise, give default values to the arguments.
- Don't forget to include comments to document the arguments of the function.

4.2 Using `annuity()`

Use your function `annuity()` to find how much money Mrs. Savingswood will have at the end of 15 years.